Class 17: Private Equity

April 2, 2026

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Today's Question

Yale's endowment has ~40% in private equity and venture capital.
Are they geniuses, or just taking risks they don't fully see?

Positioning

Where We've Been	Where We Are	Where We're Going
Evaluating public market managers	Illiquid alternatives: PE/VC	Hedge funds

Learning Objectives

By the end of today's class, you should be able to:

1. Describe the PE/VC fund structure (LP/GP, carried interest, J-curve)
2. Explain why IRR can be misleading for PE performance
3. Articulate the debate over PE "alpha" vs. leverage + illiquidity premium

Alternative Investments Introduction

Let's broaden the scope of the investable universe:

- Private equity, hedge funds, fixed income, and derivatives

Everything we learned in the first half of class applies, but...

- Data is more opaque
- Liquidity is a bigger issue
- Institutional details matter more
  - E.g. lock-up periods, redemption, fee structure
- Efficiency (maybe?)

What Is Private Equity?

A PE fund is a financial intermediary typically formed as a limited partnership:

- **General Partner (GP):** Manages funds, makes investments
- **Limited Partners (LPs):** Passive investors (pension funds, endowments)

A PE fund invests only in private (or soon-to-be-private) companies:

- **Costs:** information asymmetry and illiquidity
- **Benefits:** control

Private Equity: Structure

A PE fund takes an **active** role in the firm:

- Added value through management/monitoring

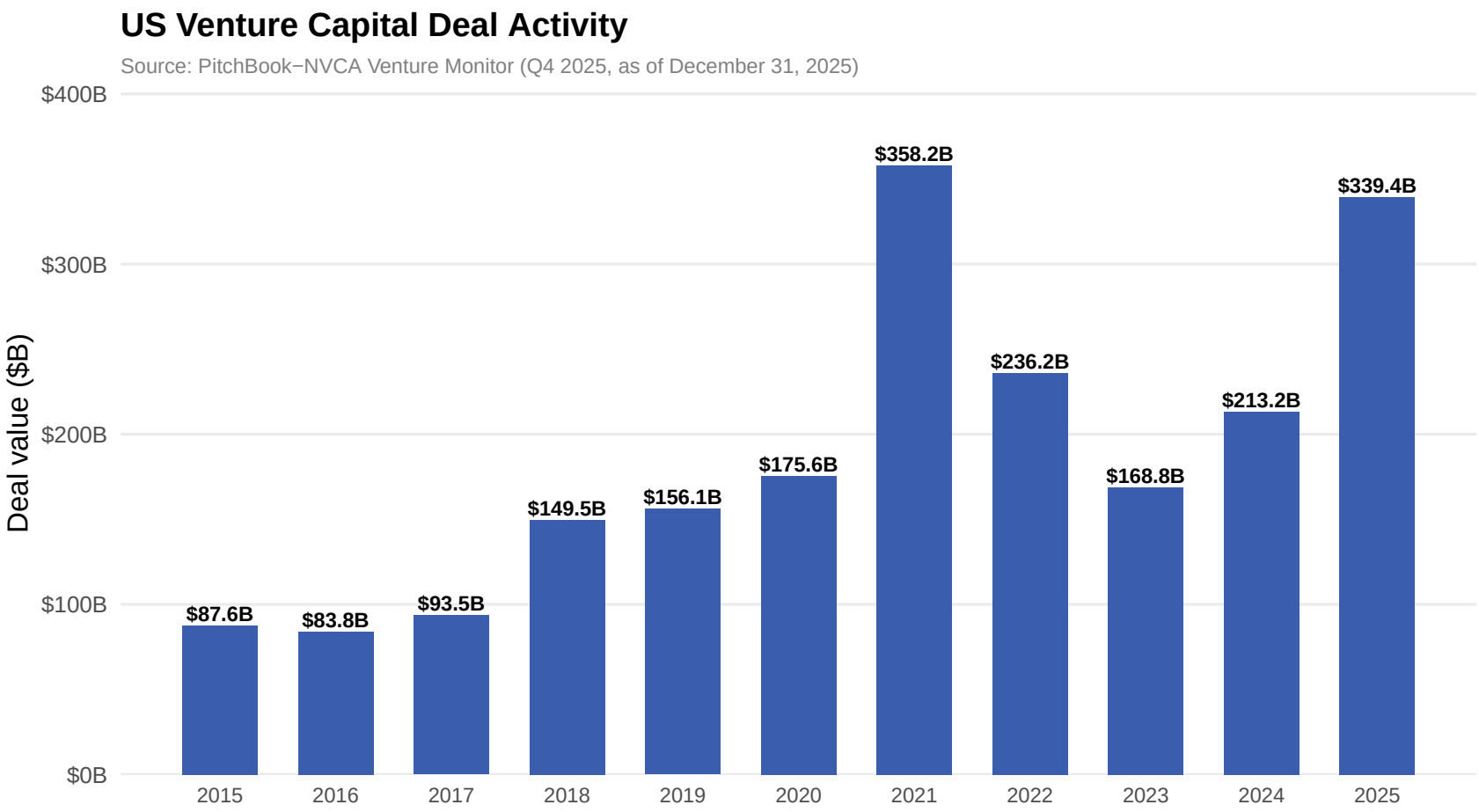
A PE fund has an **exit strategy:**

- Because of illiquidity of investment, targets must provide a clear path to exit (e.g. IPO or sale to another company)
- Funds are closed-end with lifespans of 10–12 years:
  - Commitments raised
  - Capital calls/investments made
  - “Harvesting” period

Table 2: Venture Capital-Backed Companies by CSA, 1975 - 2005

	Portfolio Company Location		Share of Investments in CSA		
CSA	Share of Total	#	Main Office	Branch Office	Outside
San Jose/San Francisco, CA	29.01%	4,063	56.55%	16.40%	27.04%
Boston, MA	11.67%	1,634	42.34%	8.07%	49.59%
New York, NY	8.74%	1,224	47.94%	2.37%	49.69%
Los Angeles, CA	6.08%	851	11.93%	2.53%	85.54%
Washington, DC	4.17%	584	20.96%	6.37%	72.67%
San Diego, CA	3.53%	494	6.71%	3.75%	89.55%
Dallas, TX	2.93%	411	17.04%	9.25%	73.71%
Seattle, WA	2.73%	383	17.40%	0.25%	82.35%
Denver, CO	2.63%	369	22.68	0.55%	76.78%
Atlanta, GA	2.48%	348	20.50%	0.33%	79.17%
Chicago, IL	2.16%	303	30.70%	0.85%	68.44%
Philadelphia, PA	3.16%	302	12.91%	2.00%	85.09%
Other	21.70%	3,040	16.41%	1.19%	82.40%
Total	100%	14,006	35.63%	7.83%	56.54%

Venture Capital: Deal Activity

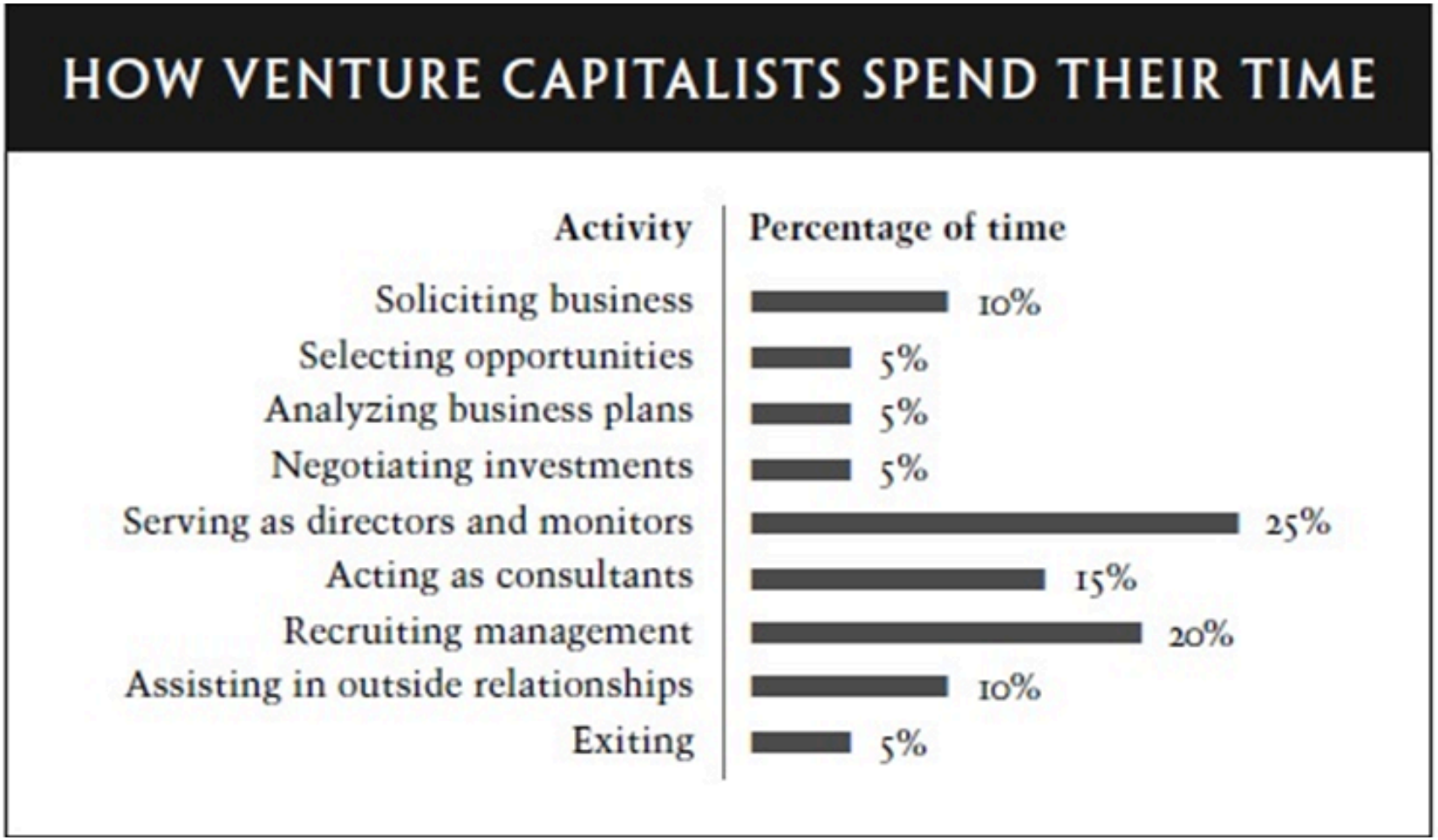


What Value Does Venture Capital Add?

1. **Screening:**
   - VCs as winner-pickers — select 1-in-400 deals to fund
   - Do VCs bet the jockey or the horse?
   - High management turnover suggests the horse
2. **Advising/Monitoring:**
   - Requires control rights
   - Investments typically include voting rights, board seats, esp. in bad times
3. **Exit**

Important to have good VCs for good returns!

How VCs Spend Their Time



How Do Venture Capitalists Make Decisions?

Gompers et al. (2020) find that:

1. When selecting deals, VCs place the greatest importance on the **management team**
2. VCs devote substantial resources to deal sourcing, selection, and post-investment value-add
3. Most VCs **do not use DCF or NPV** techniques to evaluate investments. They rely more on valuation multiples and IRR
4. After investing, VCs are **actively involved** with their portfolio companies
5. VCs are generally inflexible on key economic terms in contracts like liquidation preferences, anti-dilution rights, and vesting
6. VC firms are small, with only 14 employees on average, half of whom are senior investment professionals. Most of their time is spent on sourcing and working with portfolio companies

Buyout Funds

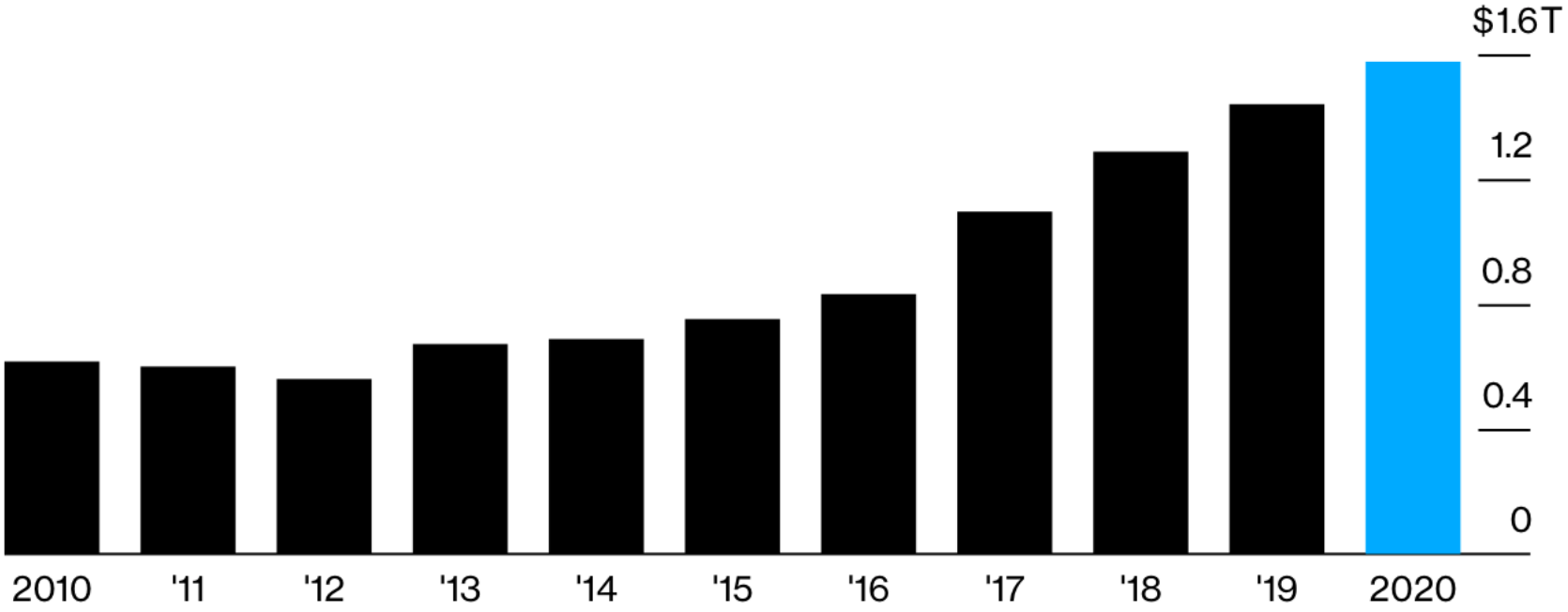
In contrast to VC, (Leveraged) Buyout (LBO) funds focus on **established firms**

- Firms are either private to begin with, or are taken private (as in a management buyout)
- Large amounts of **new debt** are used to finance the repurchase of existing equity
  - Often 90/10 debt-to-equity ratio in new company
  - Example: RJR Nabisco, Dell

There's big demand: in 2020, LBO funds had $1.6 trillion to spend on buyouts

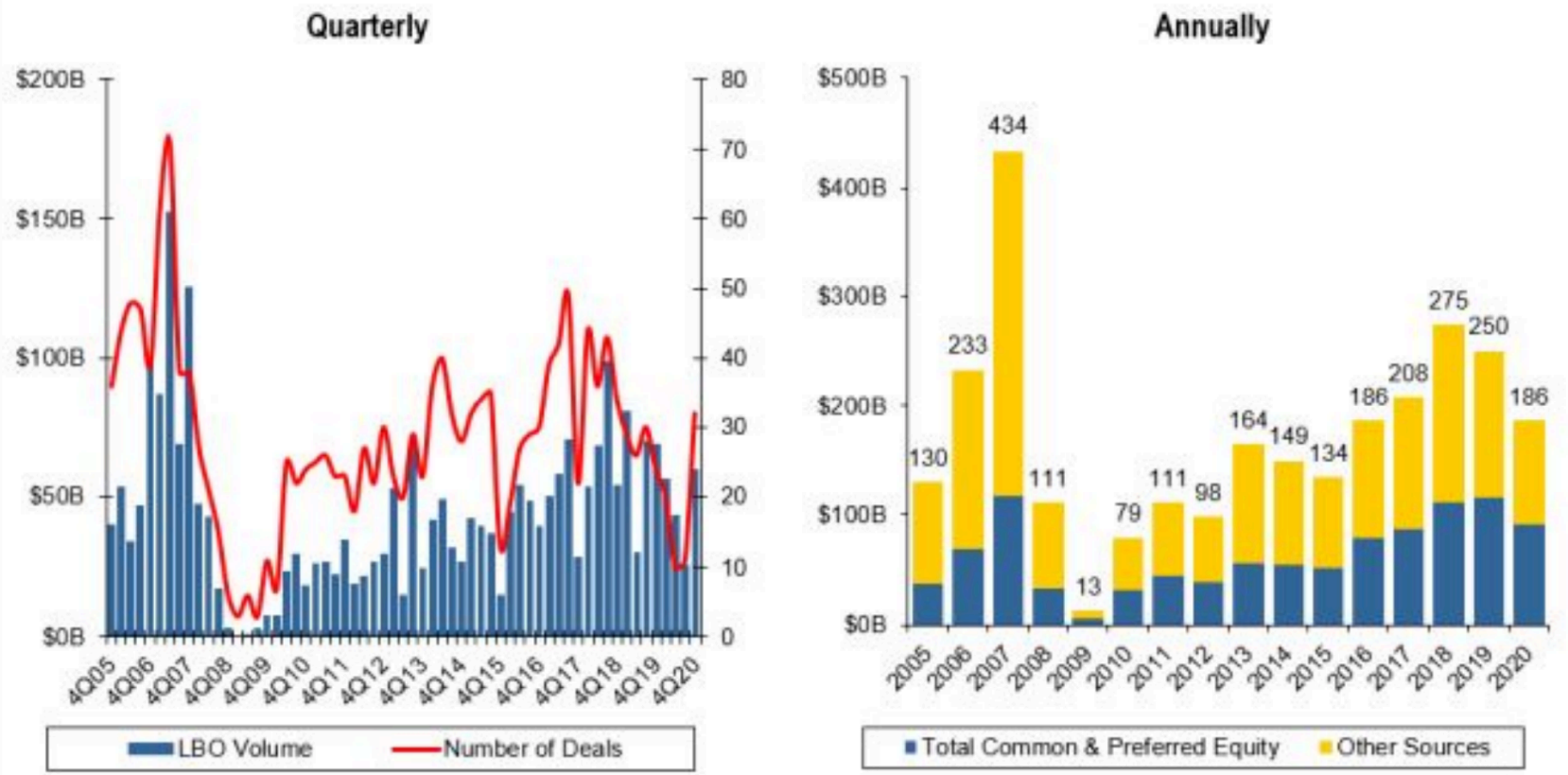
LBO War Chest

Private equity's dry powder tops $1.5 trillion



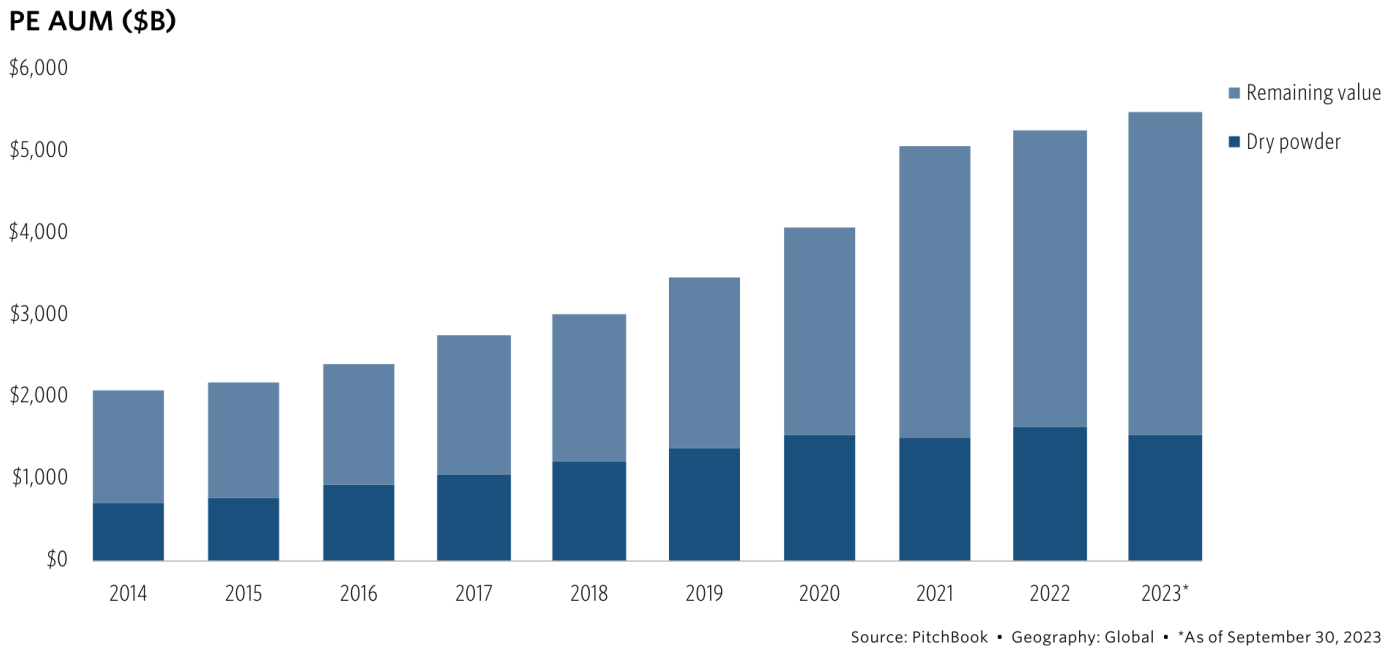
Source: Preqin
Note: 2020 as of November

LBO Capital Structure



Data through Dec. 31, 2020.

PE Assets Under Management

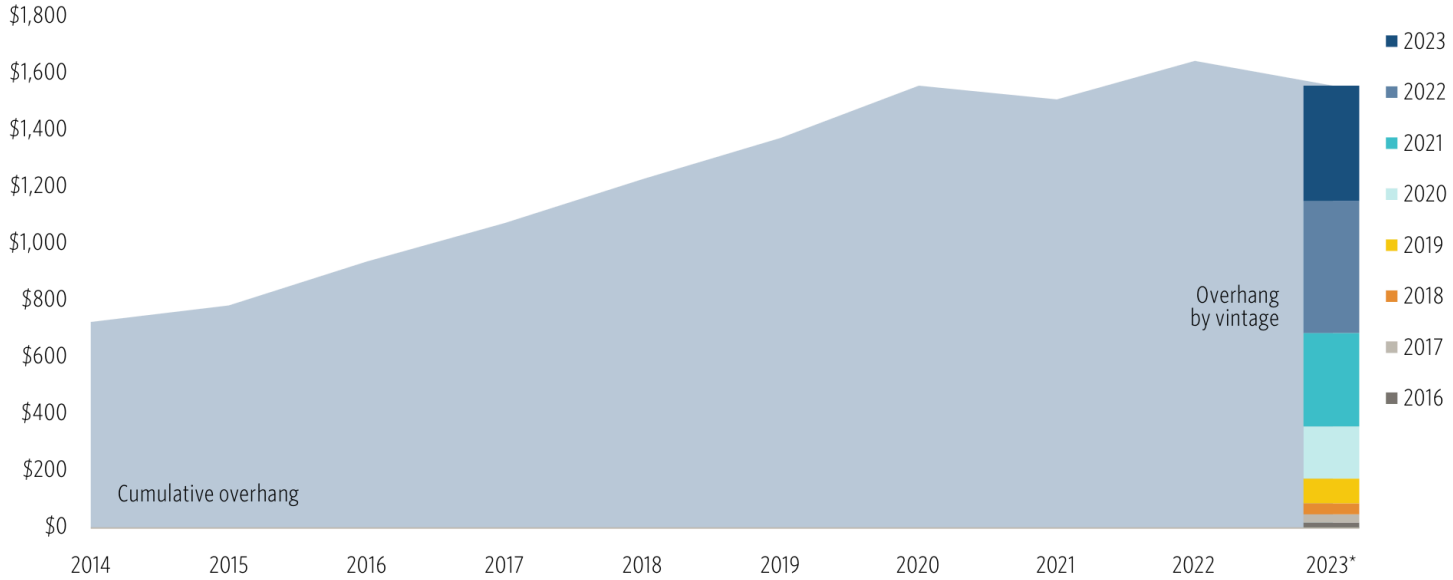


Source: PitchBook, Q1 2024 Global Private Market Fundraising Report (as of Sept. 30, 2023)

PE Dry Powder by Vintage

PitchBook

PE dry powder ($B) by vintage



Source: PitchBook • Geography: Global • *As of September 30, 2023

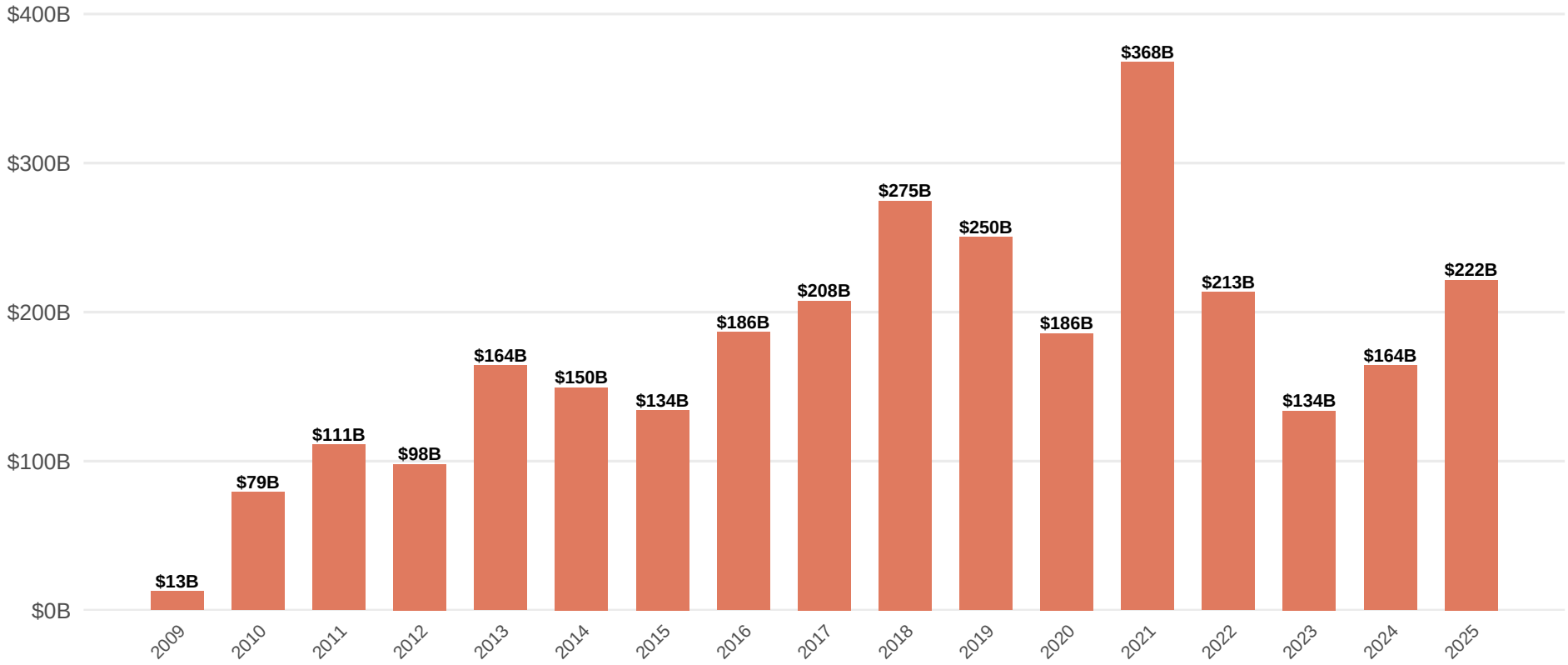
PE AUM ($B)

Source: PitchBook, Q1 2024 Global Private Market Fundraising Report (as of Sept. 30, 2023)

LBO Deal Activity

Total US LBO Transaction Volume

Includes all sources (loans, secured/unsecured debt, equity)
Source: PitchBook LCD, Q4 2025 US LBO Debt Report



Buyout Funds: Value-Add

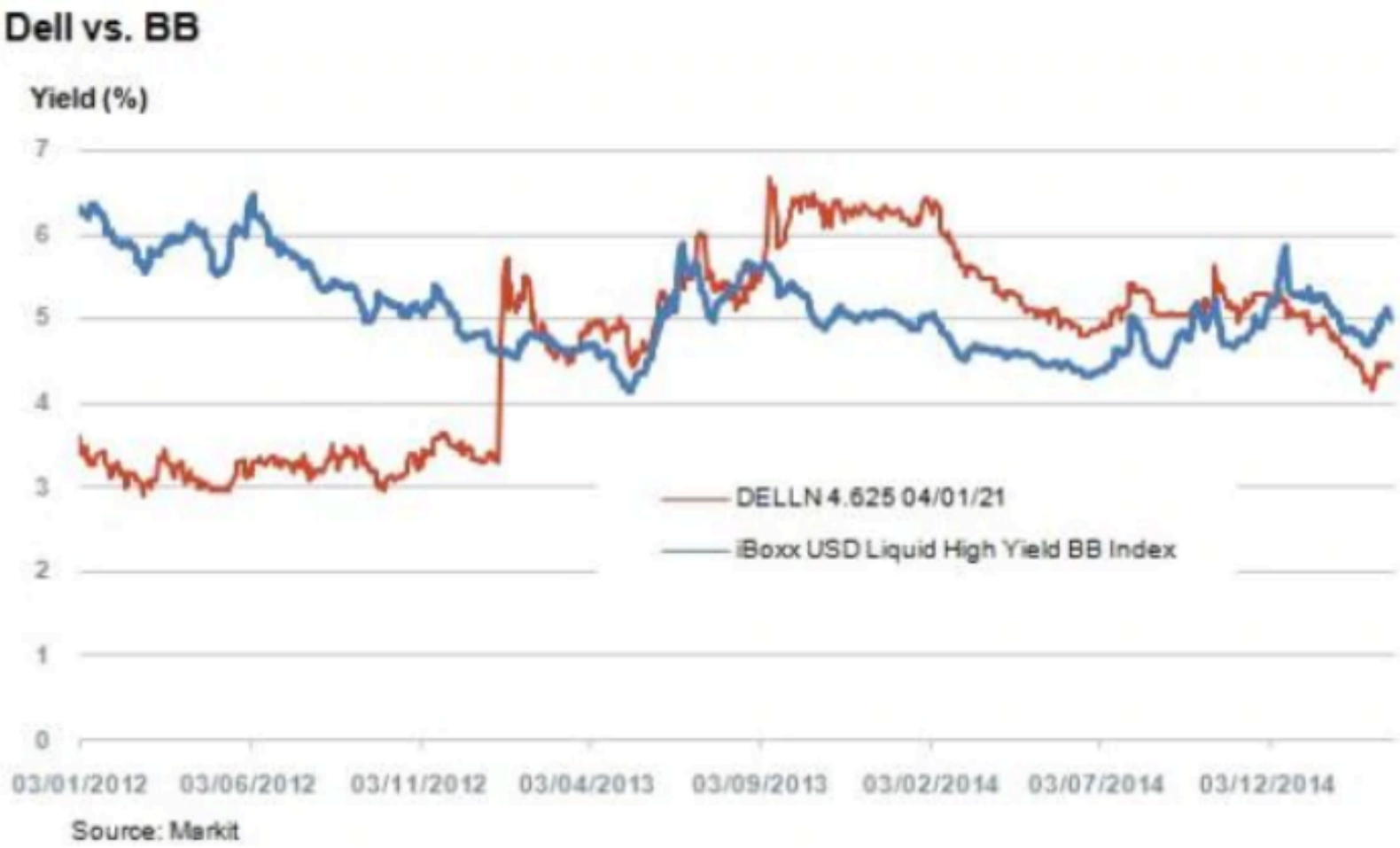
Where does the value-add come from?

- No need to take a company private to exploit an undervalued stock
- Similar to VC, **control must matter**

What types of things do LBO firms do once they have control?

- Spin-offs of conglomerates
  - Beatrice Foods
- Efficiency gains (cost-cutting, union negotiations, layoffs)
  - Safeway

Or Is It Just Leverage?



Private Equity Fund Contracts

Incentives of GPs and LPs may not always be aligned

GPs may have incentives to shirk or take excessive risk

Fees, co-investment by fund managers, and covenants discipline GPs and protect the LPs:

- **Fees:** flat rate (1–3%) plus 20–30% performance fee (carried interest)
- **Co-investment:** GPs often required to personally invest in the fund (typically 1%)
- **Covenants:** restrictions on activities

Private Equity Fund Covenants

THE NUMBER OF COVENANTS IN 140 VENTURE PARTERNSHIP AGREEMENTS, BY YEAR

	PERCENTAGE OF CONTRACTS WITH COVENANT IN:		
	1978–82	1983–87	1988–92
Covenants relating to the management of the fund:			
Restrictions on size of investment in any one firm	33.3	47.1	77.8
Restrictions on use of debt by partnership	66.7	72.1	95.6
Restrictions on coinvestment by organization's earlier or later funds	40.7	29.4	62.2
Restrictions on reinvestment of partnership's capital gains	3.7	17.6	35.6
Covenants relating to the activities of the general partners:			
Restrictions on coinvestment by general partners	81.5	66.2	77.8
Restrictions on sale of partnership interests by general partners	74.1	54.4	51.1
Restrictions on fund-raising by general partners	51.9	42.6	84.4
Restrictions on other actions by general partners	22.2	16.2	13.3
Restrictions on addition of general partners	29.6	35.3	26.7
Covenants relating to the types of investment:			
Restrictions on investments in other venture funds	3.7	22.1	62.2
Restrictions on investments in public securities	22.2	17.6	66.7
Restrictions on investments in leveraged buyouts	.0	8.8	60.0
Restrictions on investments in foreign securities	.0	7.4	44.4
Restrictions on investments in other asset classes	11.1	16.2	31.1
Total number of partnership agreements in sample	27	68	45
Average number of covenant classes	4.4	4.5	7.9
Average number of covenant classes (weighted by fund size)	4.4	4.6	8.4

The Challenge with Evaluating Private Equity

Private equity is, by definition, private — limited trading with no continuously posted price. How can we evaluate the risk-return profile?

Two main (and limited) ways:

1. **Multiples on investment**
2. **IRR (discounted cash flows)**

Multiples: the value out divided by the initial payment in

$$M = \frac{\sum_{t=1}^{T} C_t}{P_0}$$

IRR: the rate that sets the PV of all cash flows to zero

$$0 = P_0 + \sum_{t=1}^{T} \frac{C_t}{(1+r)^t}$$

Private Equity Performance

	Equal Weighted			Size Weighted		
Sample	All Funds	VC Funds	Buyout Funds	All Funds	VC Funds	Buyout Funds
IRR_{VE5}	0.11	0.10	0.12	0.13	0.15	0.12
	0.18	0.20	0.14	0.18	0.30	0.13
	(0.36)	(0.40)	(0.24)	(0.32)	(0.46)	(0.23)
	[0.02;0.22]	[0.02;0.21]	[0.02;0.22]	[0.03;0.23]	[0.04;0.42]	[0.02;0.21]

- Table reports mean, median, standard deviations, and 25th–75th percentile of dollar-weighted returns (IRR) to PE funds
  - Why dollar-weighted returns?
  - On average, roughly equivalent to S&P 500 during same period (net of fees)

Comparing to the Market: PME

The comparable results prompt a simple comparison: what is the return of the contribution to the fund if it had been put in the market?

This is the **Private Market Equivalent (PME):** the funds placed into PE are evaluated based on the opportunity cost of missing out on the market

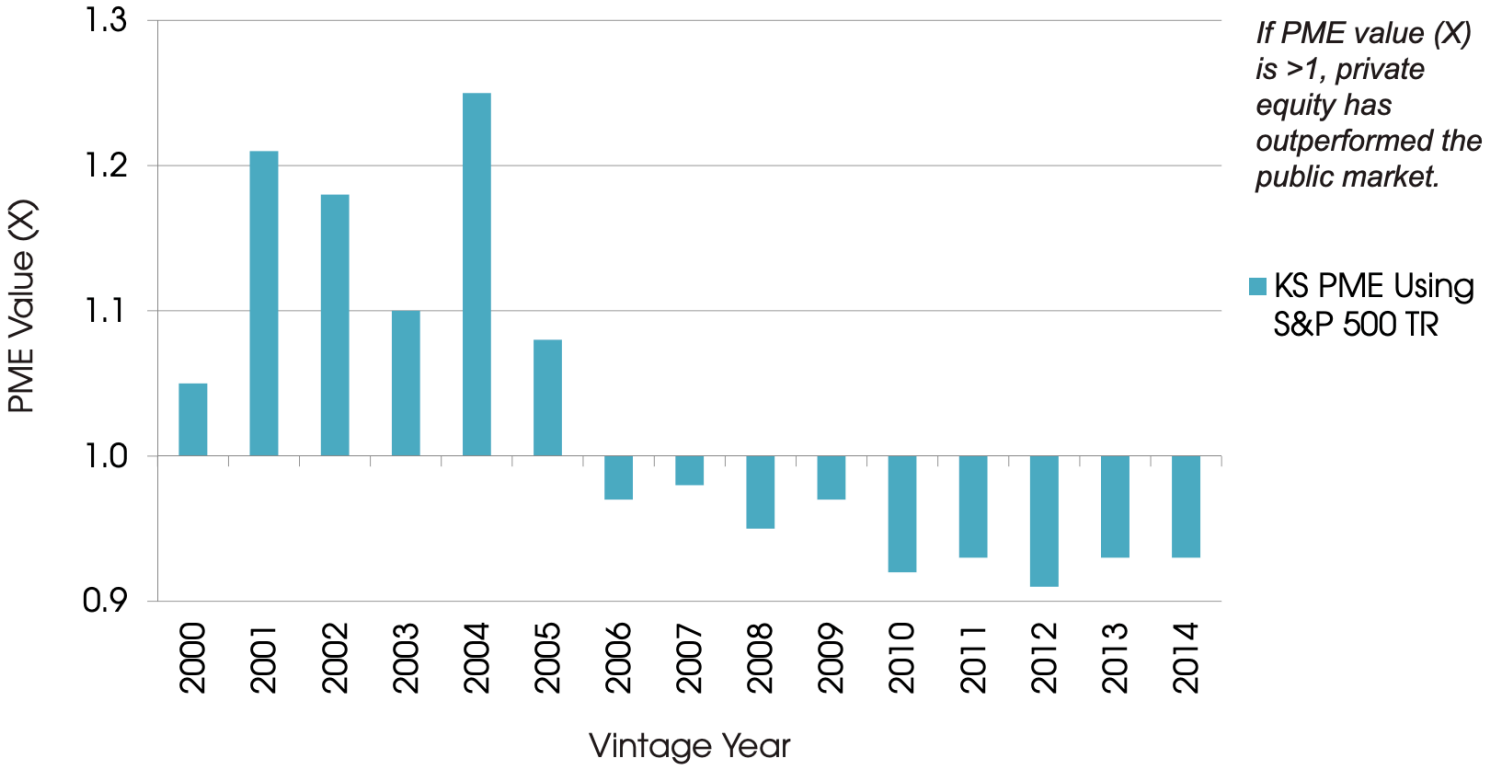
This approach can be much better than just IRR or multiples:

- A rising tide lifts all boats
- Effectively adjusts payouts based on the return in the market
- **But...** assumes a β of 1!

With PME, IRR, and multiples, can compare the relative return on investments (but not riskiness)

PE Returns by Vintage Year

Fig. 3: KS PME: All Private Equity - All Regions (As of 31 December 2014)



Source: Preqin Performance Analyst

Recent Returns: PE and VC in Context

Cambridge Associates US PE/VC Benchmark (as of June 30, 2024):

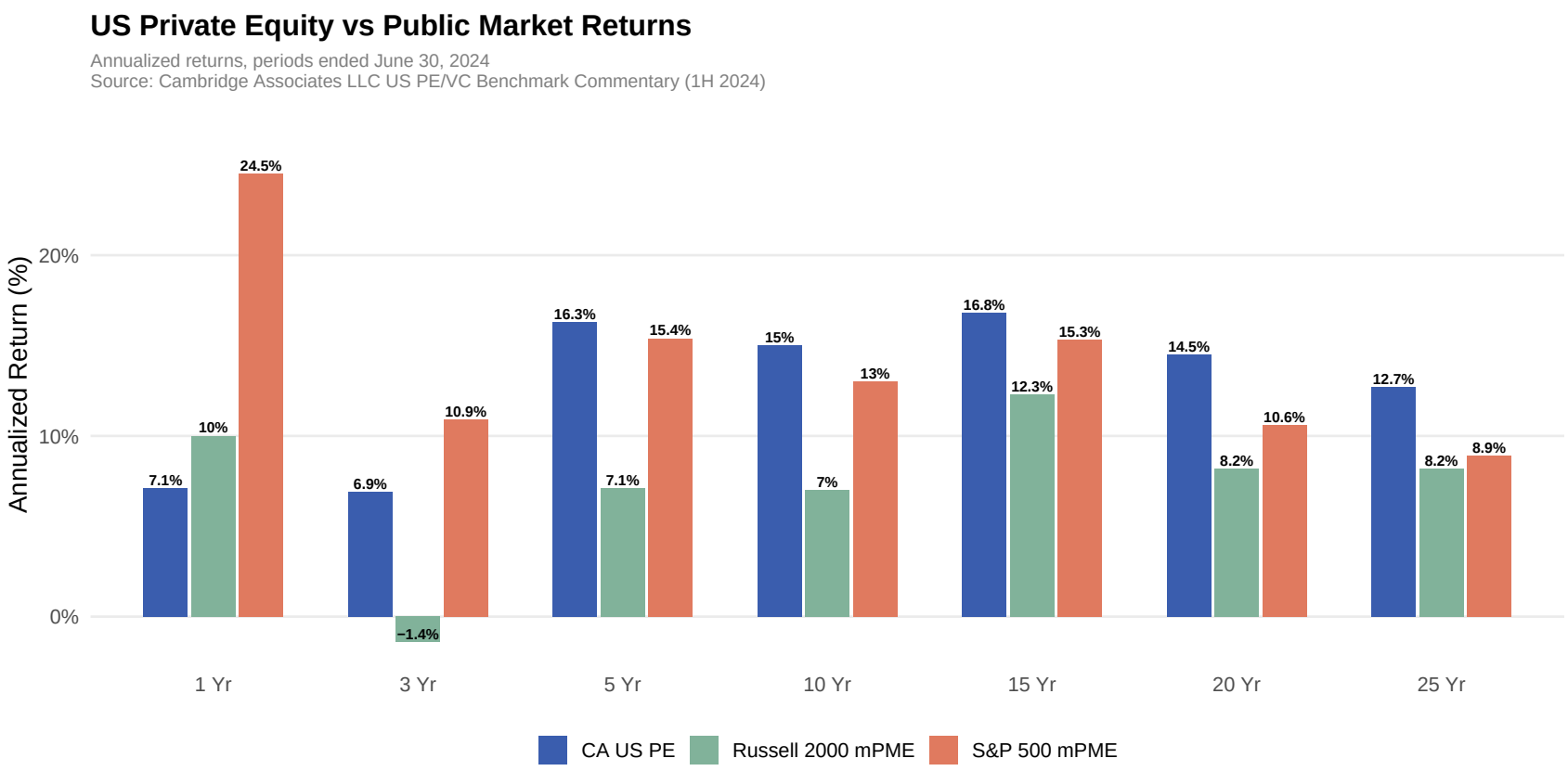
- In 1H 2024, US PE earned **3.4%** and US VC earned **1.4%**
- Both trailed the S&P 500 (+15.3%) and Nasdaq (+18.6%) over the same period
- Large-cap IT dominance has driven public indexes far ahead over 1–3 year horizons

Key insight:

- Over **longer horizons** (10–25 years), PE and VC have outperformed public markets
- But the **recent gap** is striking — raises questions about the illiquidity premium

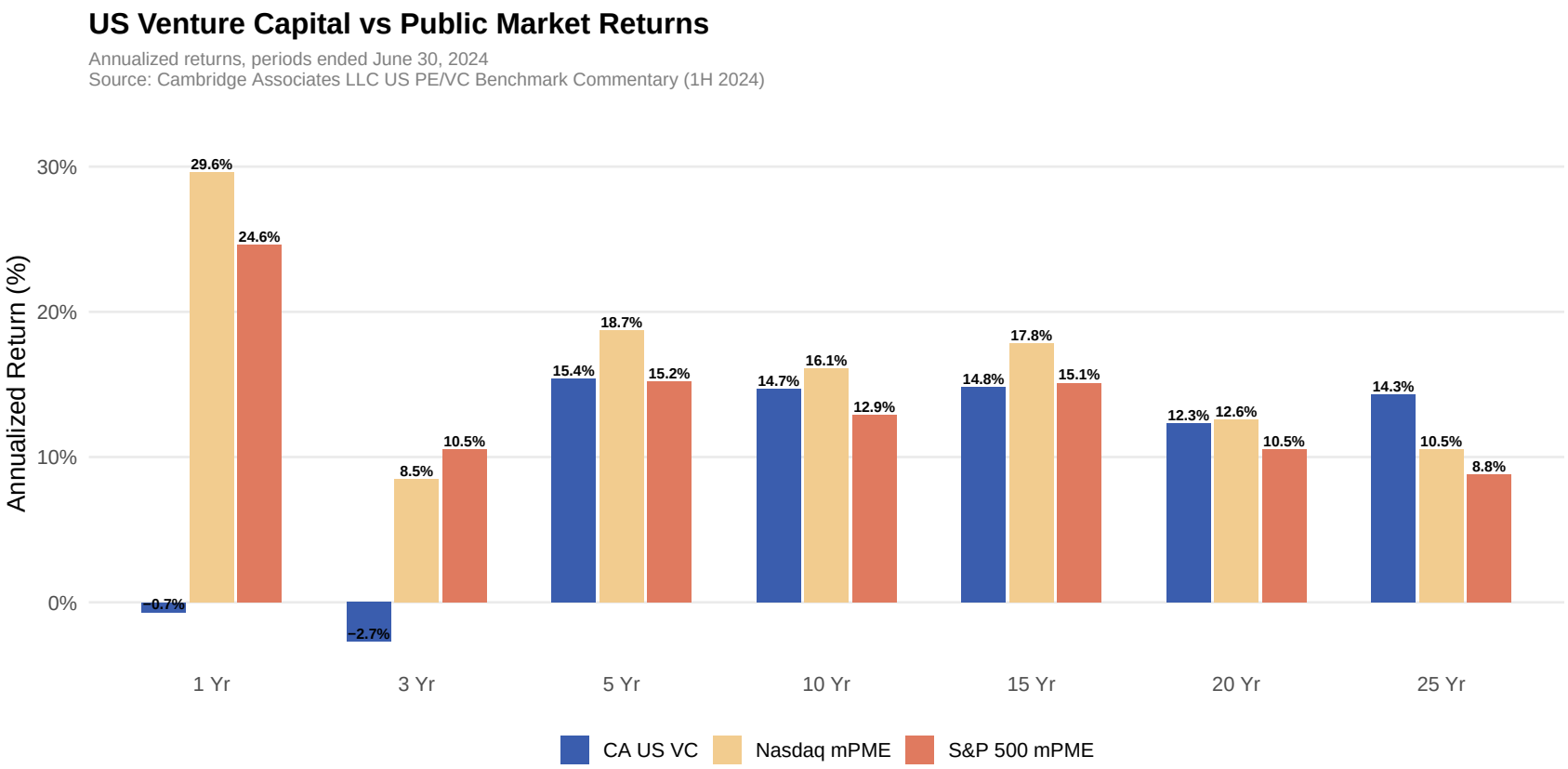
Source: Cambridge Associates LLC US PE/VC Benchmark Commentary, 1H 2024 Data (Jan. 2025)

US Private Equity vs Public Markets



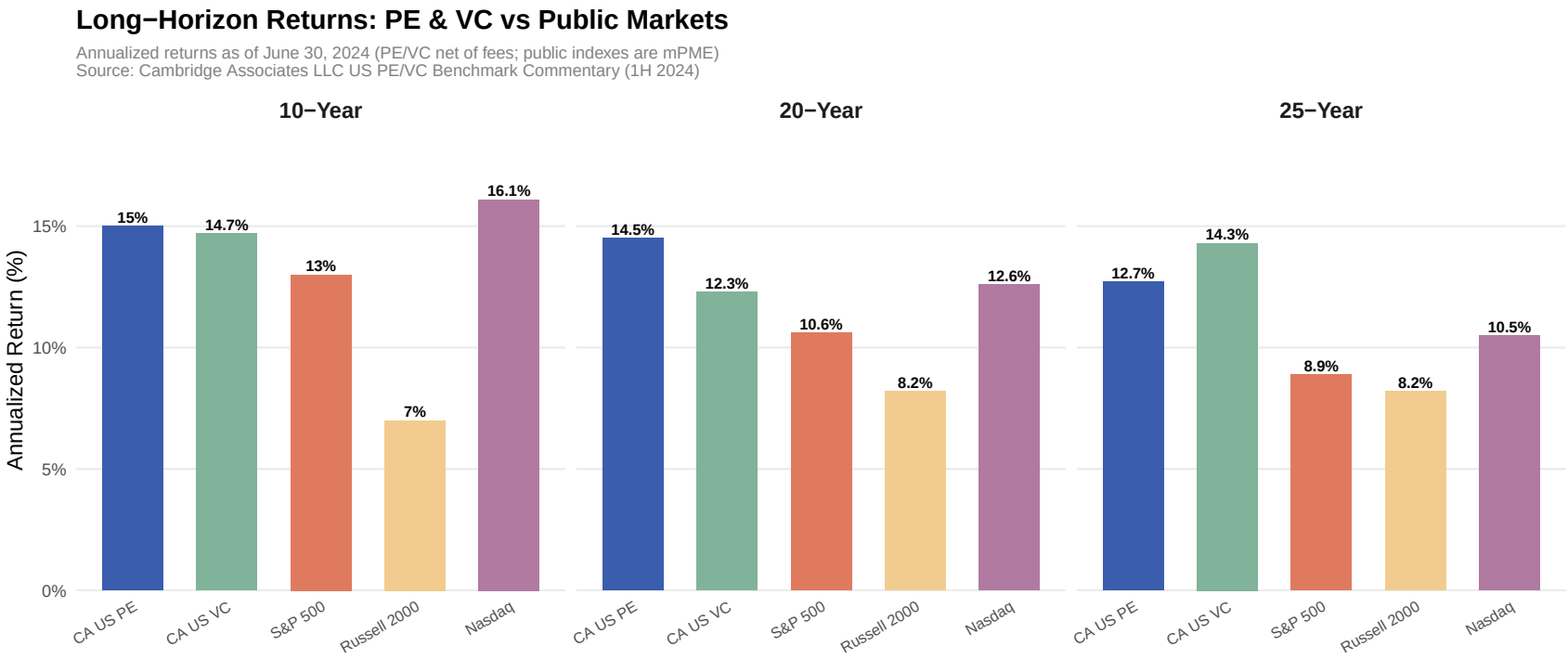
Source: Cambridge Associates LLC US PE/VC Benchmark Commentary (1H 2024). Returns annualized, periods ended June 30, 2024. PE returns net of fees; public indexes are mPME.

US Venture Capital vs Public Markets



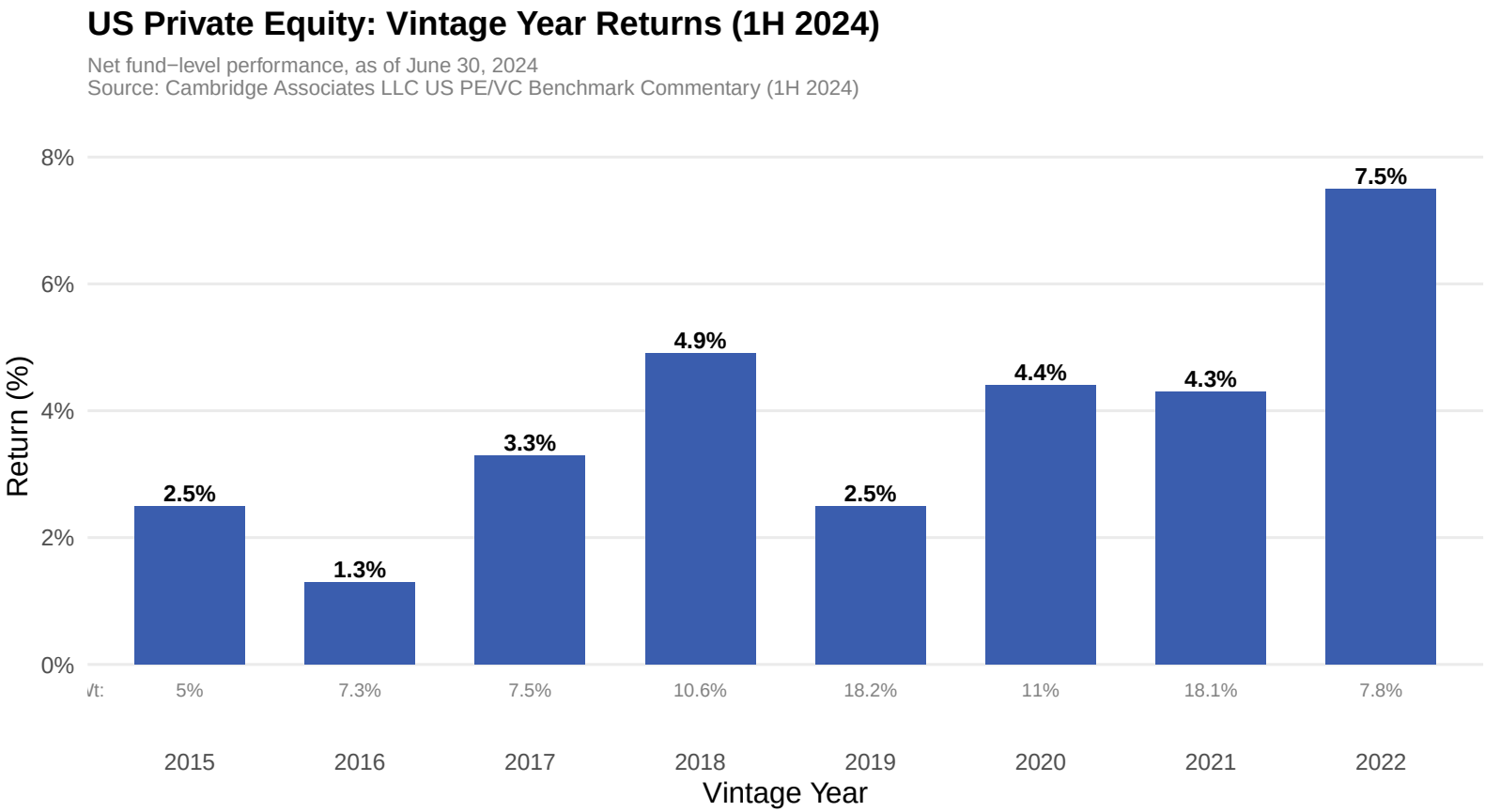
Source: Cambridge Associates LLC US PE/VC Benchmark Commentary (1H 2024). Returns annualized, periods ended June 30, 2024. VC returns net of fees; public indexes are mPME.

Long-Horizon Returns: PE & VC vs Public Markets



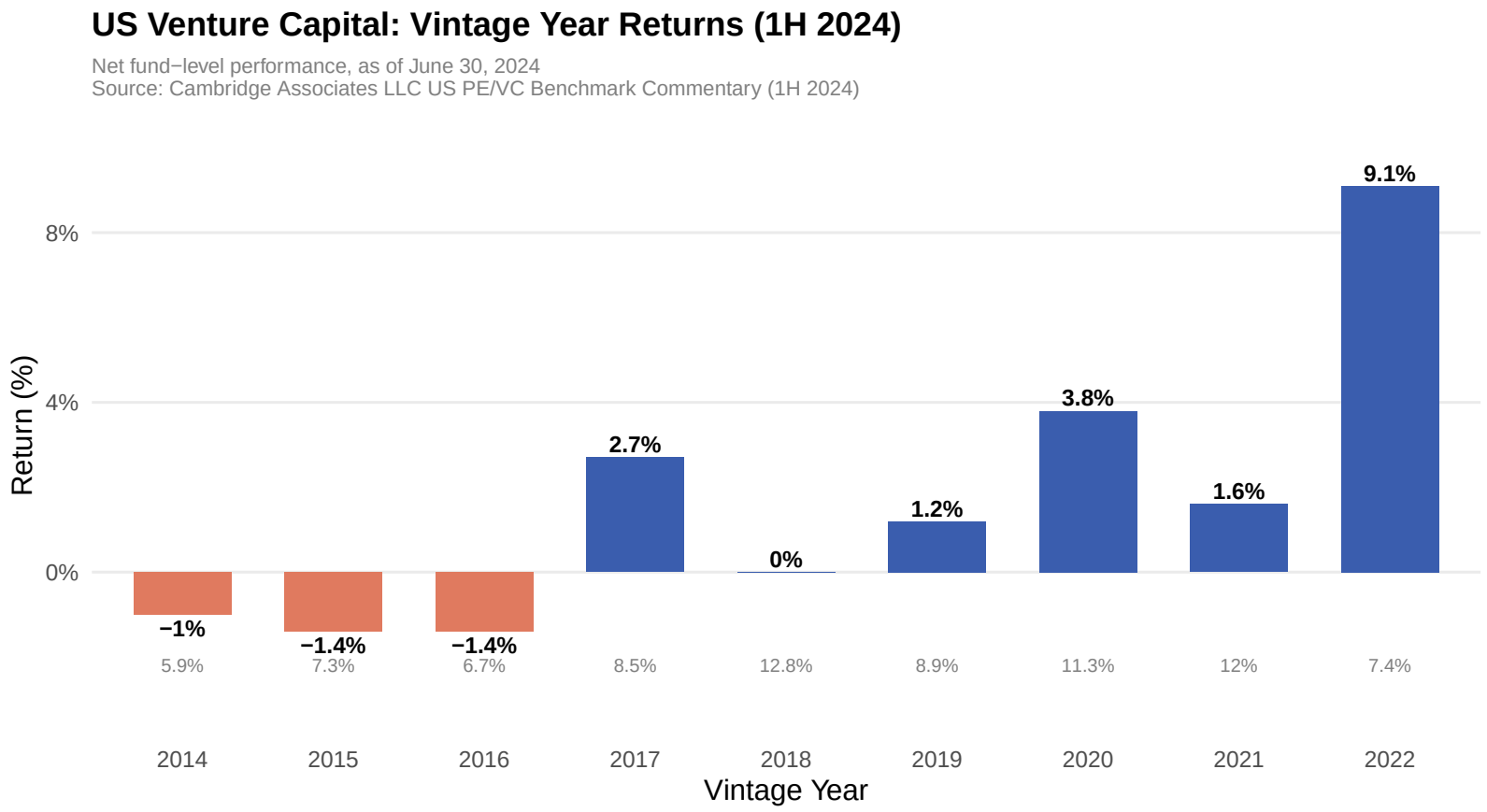
Source: Cambridge Associates LLC US PE/VC Benchmark Commentary (1H 2024). PE/VC returns net of fees; public indexes are mPME.

PE Vintage Year Returns (1H 2024)



Source: Cambridge Associates LLC. Net fund-level performance as of June 30, 2024. Weights show % of index value.

VC Vintage Year Returns (1H 2024)



Source: Cambridge Associates LLC. Net fund-level performance as of June 30, 2024. Weights show % of index value.

What Should We Make of Recent PE/VC Returns?

The recent underperformance raises important questions:

1. **Is the illiquidity premium shrinking?**
   - More capital chasing deals → compressed returns
   - PE AUM has grown enormously ($1.4T in CA's index alone)
2. **Are public markets just unusually strong?**
   - Magnificent 7 / large-cap IT drove most of the S&P 500 and Nasdaq gains
   - PE/VC are more like small/mid-cap and are less IT-concentrated than Nasdaq
3. **Measurement issues**
   - PE valuations are **stale** — marked quarterly by GPs, not the market
   - VC index was down in 8 of 10 quarters since early 2022
   - Distributions have lagged capital calls since 2022

The long-run story still favors PE/VC, but the recent data is humbling

How to Evaluate the Riskiness of PE Investments?

PME, IRR, and multiples give ways to evaluate PE **funds**. But what about riskiness?

Two approaches:

1. Consider the variance in the time series. Subject to serious data issues (selection bias)
2. Simulations — model capital calls / cash flows

Neither of these methods adjust for aggregate risk factors.

Gupta and Van Nieuwerburgh (2020) argue that PE is actually quite exposed to other factors:

- Model the cash flows and compare to public market factors
- Highly exposed to SMB, HML, and other public equities
- *Suggestive that you could get the same return by investing in public equities instead (at least on average)*

How to Identify Good PE Funds?

Pick past winners!

Returns to private equity are **highly persistent.** (Why?)

Issue: everyone wants to get in funds with strong performance

- Who gets access?
- What incentive effects does this create?

PE Performance: Persistence

General Partner (PE fund) performance is persistent:

PME_{t-1}	0.54	0.49			0.46
	(0.17)	(0.21)			(0.21)
PME_{t-2}		0.28	0.39		0.28
		(0.13)	(0.14)		(0.13)
PME_{t-3}				0.32	
				(0.24)	
log(Size)					0.09
					(0.05)
log(Sequence)					0.20
					(0.20)

Private Equity: Other Issues

- Often the binding constraint for successful PE funds is **investment opportunities**, not investors
- Leads to the **money-chasing-deals** phenomenon
- During periods of high flows into private equity, funds pay higher valuations for their acquisitions

Next Class: April 7

Topics: Hedge Funds I

- Hedge fund strategies
- Pairs trading
- Merger arbitrage